LCI INTERSECTION MULTIPLICITY AS LEADING COEFFICIENT OF A HILBERT POLYNOMIAL

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0B03: sources/stacks-chow/tags.txt, source lines 13846–13846. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00KQ: sources/stacks-chow/tags.txt, source lines 760–760. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00P0: sources/stacks-chow/tags.txt, source lines 914–914. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 063I: sources/stacks-chow/tags.txt, source lines 7832–7832. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0AZN: sources/stacks-chow/tags.txt, source lines 13831–13831. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00P1: sources/stacks-chow/tags.txt, source lines 915–915. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a nonsingular variety. Let $W, V \subset X$ be closed subvarieties which intersect properly. Let Z be an irreducible component of $V \cap W$ with generic point ξ. Suppose the ideal of V in $\mathcal{O}_{X,\xi}$ is cut out by a regular sequence $f_1, \ldots, f_c \in \mathcal{O}_{X,\xi}$. Then $e(X, V \cdot W, Z)$ is equal to $c!$ times the leading coefficient in the Hilbert polynomial*

$$t \mapsto length_{\mathcal{O}_{X,\xi}} \mathcal{O}_{W,\xi}/(f_1, \ldots, f_c)^t, \quad t \gg 0.$$

In particular, this coefficient is > 0.

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Proof. The equality

$$e(X, V \cdot W, Z) = e_{(f_1,\ldots,f_c)}(\mathcal{O}_{W,\xi}, c)$$

follows from the more general Lemma 0B03. To see that $e_{(f_1,\ldots,f_c)}(\mathcal{O}_{W,\xi}, c)$ is > 0 or equivalently that $e_{(f_1,\ldots,f_c)}(\mathcal{O}_{W,\xi}, c)$ is the leading coefficient of the Hilbert polynomial it suffices to show that the dimension of $\mathcal{O}_{W,\xi}$ is c, because the degree of the Hilbert polynomial is equal to the dimension by Algebra, Proposition 00KQ. Say $\dim(V) = r$, $\dim(W) = s$, and $\dim(X) = n$. Then $\dim(Z) = r + s - n$ as the intersection is proper. Thus the transcendence degree of $\kappa(\xi)$ over $\mathbf{C}$ is $r + s - n$, see Algebra, Lemma 00P0. We have $r + c = n$ because V is cut out by a regular sequence in a neighbourhood of ξ, see Divisors, Lemma 063I and then Lemma 0AZN applies (for example). Thus

$$\dim(\mathcal{O}_{W,\xi}) = s - (r + s - n) = s - ((n - c) + s - n) = c$$

the first equality by Algebra, Lemma 00P1. □